

For my improvement, it seems as if genre is my ground truth data, so I felt that filtering recommendation candidates to the query song's genre before ranking by cosine similarity would be best. A problem with this of course is the song recommender not recommending actually similar songs from a different genre, but it's obviously more likely that listeners would enjoy songs from the same genre.

Because it was impossible for a song of a different genre to be recommended, the precision@3 that I got was exactly 1. As for the dot product, cosine similarity, and tfidf cosine similarity's precision@3, they were 0.092, .191, and .2101 respectively. The dot product being the worst made sense, as throughout the lab it gave the poorest results for song recommendations due to being heavily biased towards long lyric heavy songs. The non-tfidf cosine was roughly twice as effective as the dot product which is consistent with my earlier findings, as it removes the magnitude bias. Finally, the tfidf cosine turned out to be the best, as the removal of very common words helped narrow down similarities between songs.

However, a precision@3 of .21 is still quite low, moreover genre is a poor gauge of what songs should be recommended. Overall, filtering by genre first obviously led to exclusively same genre songs being recommended, however this isn't necessarily a good thing and there are definitely other filters that should be tried out.